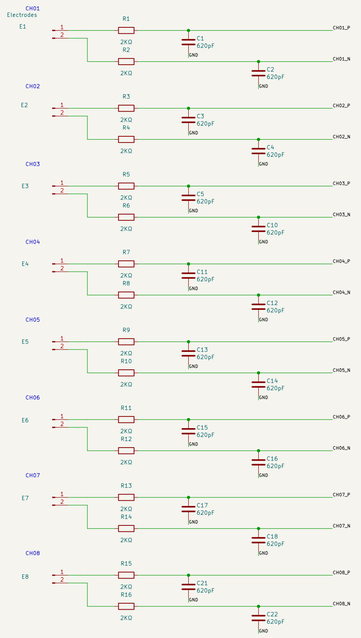
OPEN-sEMG-16 V1.4 | USB power / Wi-Fi data | 16 x 24-bit channels at target 2 kSPS

BENCH ONLY: simulated, non-human-connected signals. Never attach a person while laptop USB is connected.
USB supplies power/programming/debug; acquisition frames go over built-in Wi-Fi. Firmware and hardware remain untested.

ADS1298 #1 / channels 1-8

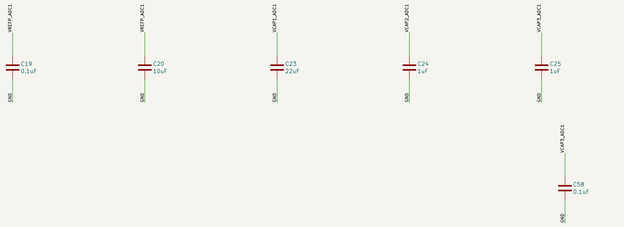




AVDD local DVDD local AVDD1 local
Route each bypass at its pins: star AVDD1/AVSS1 to AVDD/AVSS.
All three domains use the SAME DevKit 3.3V rail and common return.

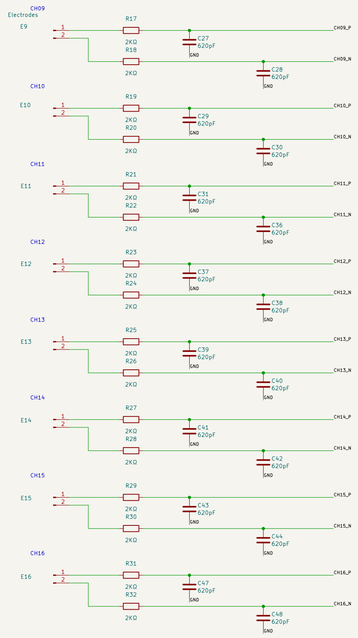


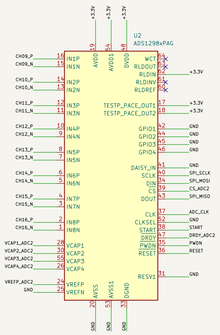
Reference / internal bypass nodes: separate per ADC; never join VCAPx.
VREFP: 10uF || 100nF to VREFN/GND: enable internal 2.4V reference.
VCAP1=22uF, VCAP2=1uF, VCAP3=1uF || 100nF, VCAP4=1uF.
VCAP3 is about5.2V on3.3V supply; >=10V rated, bias-qualified parts.



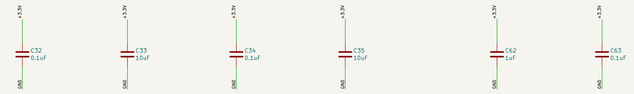
Input pairs: 2k per leg (0.1%, >=0.1W), 620pF to GND (C0G,1%).
Preserved RC topology; no added human-contact protection.
Unused PACE/RLDIN7/18/62 tied to AVDD: keep PACE/rerouting OFF.
RLDREF/RLDINV/RLDOUT/WCT unconnected: keep RLD/WCT OFF.
GPIO1-4. DAISY_IN and RESV1 grounded; configure GPIO1-4 as inputs.

ADS1298 #2 / channels 9-16

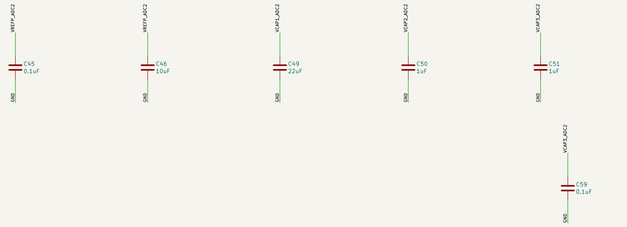




AVDD local DVDD local AVDD1 local
Route each bypass at its pins: star AVDD1/AVSS1 to AVDD/AVSS.
All three domains use the SAME DevKit 3.3V rail and common return.

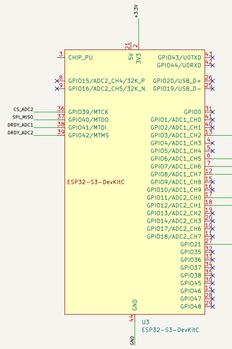


Reference / internal bypass nodes: separate per ADC; never join VCAPx.
VREFP: 10uF || 100nF to VREFN/GND: enable internal 2.4V reference.
VCAP1=22uF, VCAP2=1uF, VCAP3=1uF || 100nF, VCAP4=1uF.
VCAP3 is about5.2V on3.3V supply; >=10V rated, bias-qualified parts.



Input pairs: 2k per leg (0.1%, >=0.1W), 620pF to GND (C0G,1%).
Preserved RC topology; no added human-contact protection.
Unused PACE/RLDIN17/18/62 tied to AVDD: keep PACE/rerouting OFF.
RLDREF/RLDINV/RLDOUT/WCT unconnected: keep RLD/WCT OFF.
GPIO1-4. DAISY_IN and RESV1 grounded; configure GPIO1-4 as inputs.

ESP32-S3 DevKitC-1 / USB power



Use a complete stock DevKitC-1, not a bare module.
Laptop USB -> onboard regulator -> 3V3 header pins1/2.
No battery, external regulator or other3V3 source.
5V and CHIP_PU header branches have no carrier connection;
onboard USB, reset, boot and UART circuits remain in DevKit.

Shared communication / startup (F06/F09/F18)



Direct MOSI=GPIO11; SCLK=GPIO12; MISO=GPIO40.
CS1=GPIO21 / CS2=GPIO39: HIGH at boot: select ONE ADC only.
DOUT is high-Z with CS HIGH (TI9.4.2.1), No external MISO gate.
DRDY1=GPIO41 / DRDY2=GPIO42. Input-only, pulls disabled.
Use GPIO mux, disable pad JTAG; verify eFuses before assembly.
GPIO3 HIGH selects USB JTAG only in strap-controlled modes.

START=GPIO8; RESET=GPIO6; PWDN=GPIO7.
10k pulldowns retain LOW while MCU high-Z (also DIN/SCLK).
Boot/warm reset: initialize LOW forward latches; CS both HIGH.
Release PWDN: wait U1POR/VCAP1>1.1V; pulse shared RESET.
then hold RESET HIGH. SDATAC/configure U1 with CLK_EN=1.
U1CLKSEL=HIGH, U2CLKSEL=LOW: U1CLK drives both CLK pins.
Wait U2POR on valid clock; use CS2-only SPI RESET command.
Keep shared RESET HIGH, or restart entire clock initialization!
Configure both: 2.4V reference enable/settle/readback, then START.
POR >=2^18 clocks per ADC; observe TI11.1 timing;
TI low-input ramp wording vs CS pullups needs bench review.

Acquisition / wireless target (firmware required)

HR mode2kSPS nominal: CONFIG1 HR=1, DR=100, both ADCs.
2 x (24-bit status +8x24-bit data) =54 bytes/sample pair.
2k pairs/s =108kB/s:2MHz SPI takes216us of each500us.
SPI mode1, short traces; verify measured loaded logic/edges.
Example Wi-Fi:10 pairs +16B header/UDP =116.8kB/s at IP layer.
Use SPI DMA, separate Wi-Fi task, ring buffer, sequence IDs.
Bounded RF outages need buffering/retry; no lossless claim tested.

Electrical limits / bench prerequisites

Power budget NOT verified: identify fitted DevKit/regulator, USB current authorization, 3V3 droop, capacitor stability and temperature.
Reference DevKit v1.1 regulator: SGM2212.800mA rating; not a guaranteed available header current, 500mA USB has no full worst-case margin.
Wi-Fi355mA is a characterized TX peak at3.3V/25C. not a complete-system maximum. No external power source may join the3V3 rail.
No external analog source before rails are valid. Begin with ADC internal-short/test mux, then a grounded simulator near1.65V common mode.
At gain6 and2.4V reference: |VD|<0.4V and0.2+G|VD|/2<VCM<3.3-0.2-G|VD|/2. Keep each input within rails2k is not a clamp.
Direct ESP-ADS VIH/VOH limits meet only at0.8xVDD: loaded HIGH margin/startup ramps require scope checks. Keep RF returns away from ADC loops.
FOT/FOB body bias, wearer protection and isolation remain unimplemented. Bench power-up requires the checks in reviews_v1.4/implementation.md.

Simulated signals only — no human connection with laptop USB

McGill AI Lab	
Sheet: /	
File: open-semg-16-v1.4.kicad_sch	
Title: Open-sEMG-16 USB / Wi-Fi bench prototype	
Size: A1 | Date: 2026-10-28	**Rev: 1.4**
KiCad E.D.A. 10.0.7	Id: 1/1